Processing single cells inside semi-permeable capsules: a generic microfluidic workflow

A. Example and B. Placeholder (illustrative document, not a real publication)

Abstract

Semi-permeable capsules (SPCs) let small molecules and enzymes reach cells and their nucleic acids while keeping large products inside, so many single cells can be processed in bulk. Here we describe a generic workflow, paraphrased from published SPC methods, that takes adherent cells through dissociation, encapsulation, photocuring, lysis, reverse transcription and index PCR to a sequencing-ready library. All values are placeholders for illustration, not a validated recipe.

1. Introduction

Droplet microfluidics compartmentalises single cells, but each droplet supports only one reaction mix. Capsules with a hydrogel shell can be washed and exchanged into new reagents many times (Leonaviciene & Mazutis, Nucleic Acids Research 2023; Baronas et al., Science 2026). This note restates the workflow in a generic form for use as a test document.

2. Methods

2.1 Cell preparation

Adherent cells were cultured in a culture vessel at 37 °C with 5% CO2 until confluent (placeholder: 3 days, medium changed daily). The medium was removed and the cells were rinsed once with buffer (200 µL). Dissociation enzyme (100 µL) was added and the cells were incubated for 5 min at 37 °C to detach them. Quench medium (200 µL) was added and mixed gently to stop the enzyme. The cells were centrifuged at 300 g for 5 min and the supernatant was removed. The pellet was resuspended in wash buffer (500 µL) and the wash was repeated once. The suspension was passed through a 40 µm cell strainer. Cells were counted and viability was checked. Cells were resuspended in core solution at the loading density (placeholder: 2 x 10^6 cells/mL).

2.2 Capsule generation

The cell/core solution was co-flowed with the shell polymer solution (kept warm) and focused with fluorinated oil containing surfactant in a flow-focusing junction to make droplets of about 70 µm (placeholder flows: core 2 µL/min, shell 2 µL/min, oil 10 µL/min). The emulsion was collected and kept on ice for 15-30 min so that the shell gelled. The emulsion was broken by adding emulsion-breaking agent and the oil phase was removed. Capsule shells were photocured with 405 nm light (placeholder: 30 s) in the presence of photoinitiator. Capsules were washed three times in buffer to remove oil and free reagents and filtered through a 100 µm strainer to remove aggregates.

2.3 Processing inside capsules

The buffer was exchanged for lysis buffer and the capsules were incubated (placeholder time and temperature). Capsules were washed three times in wash buffer. The buffer was then exchanged for reverse transcription mix and the capsules were incubated (placeholder time and temperature), followed by two washes. The capsules were split into four equal portions, and index PCR was run on each portion with a different index primer mix (placeholder number of cycles). The cDNA was released by digesting the capsule shells with a shell-digesting enzyme (placeholder time and temperature).

2.4 Library clean-up and quality control

The library was bound to magnetic beads, washed twice with ethanol wash solution on a magnet and eluted in water. Library concentration and fragment size were measured.

3. Results

This is an illustrative document and reports no experimental results. A typical run is expected to give capsules of about 70 µm with one cell or none per capsule.

References

1. Leonaviciene G., Mazutis L. RNA cytometry of single cells using semi-permeable microcapsules. Nucleic Acids Research (2023).

2. Baronas et al. Science (2026).